

Self-Improving LLM Agents for Automated Web-Scraper Generation in Public Procurement

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ABSTRACT

Collecting public tender documents requires a bespoke web scraper for each of hundreds of incompatible e-procurement portals — a manual process that does not scale. We present **Vergabepilot.AI**, a system in which a large language model **synthesises a Python scraper per domain** inside a closed *generate–validate–execute–evaluate* loop that iteratively repairs its own code, complemented by a visual computer-use agent and a cost-aware cascade that reuses cached scrapers before invoking any model. On a benchmark of 100 URLs spanning 30+ portal types, the system retrieved documents for **97% of URLs at a total LLM cost of \$0.02**; across 10 evaluated models the smallest and cheapest model matched the largest, indicating that model scale is not the limiting factor for this task.

1 Introduction

Public procurement notices (*Vergabe*) are published across hundreds of heterogeneous portals (DTVP, NetServer, Cosinex, RIB iTWO, Subreport, ...), each with a distinct layout, access wall and document structure. Conventional practice builds and maintains a separate scraper per source; the engineering effort grows with the number of domains and becomes the dominant cost.

Contributions.

- 1 A closed-loop agent system that **synthesises and self-repairs** domain-specific scrapers via static-safety-checked, sandboxed execution.
- 2 A **cost-aware cascade** with a visual computer-use fallback that minimises LLM expenditure.
- 3 An **empirical study** over 100 URLs, 30+ portals and 10 LLMs, achieving 97% retrieval at \$0.02.

2 Methods — System Architecture

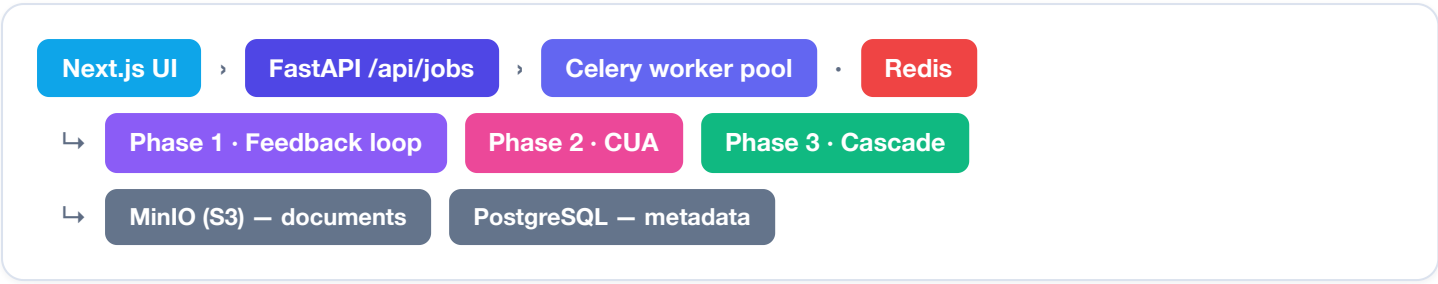


Figure 1. Jobs are submitted via the UI, enqueued and fanned out across parallel Celery workers, then routed through three extraction phases; documents persist to object storage and metadata to PostgreSQL.

3 Methods — Self-Refining Scraper Loop

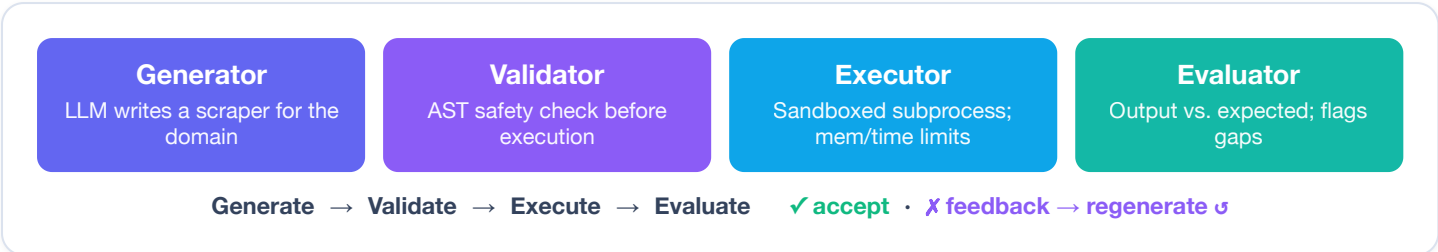


Figure 2. A self-refinement loop [1] iteratively repairs generated code until extraction succeeds — converting per-domain scraper engineering into an automated procedure.

4 Methods — Cost-Aware Cascade

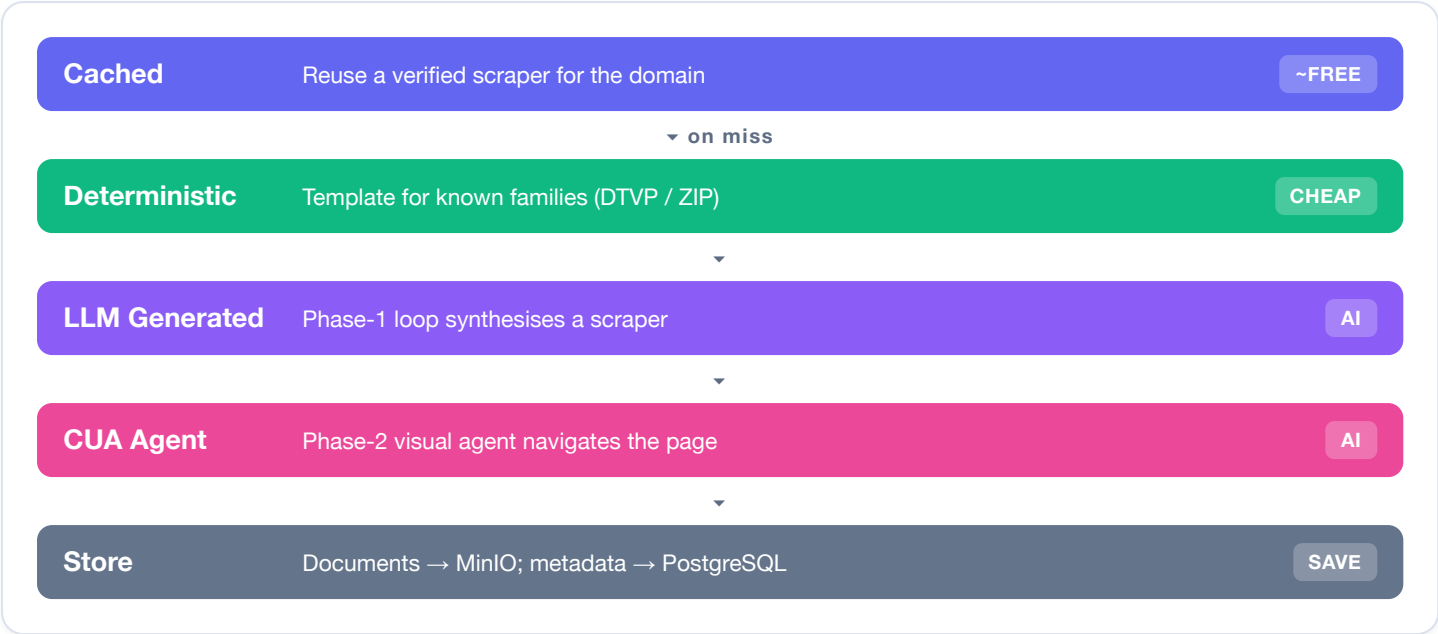


Figure 3. The cheapest viable strategy is attempted first; model invocations occur only when caching and deterministic templates miss.

5 Experimental Setup

- **Dataset:** 100 public German tender URLs spanning 30+ portal types (NetServer, DTVP, Cosinex/eVergabe 4.9, RIB, Subreport, TYPO3, SharePoint, Ariba, ...), auto-classified before any network call.
- **Models:** 10 LLMs evaluated (Gemini 2.5 Flash Lite/Flash/Pro, GPT-4o mini, GPT-4.1(/mini), Claude Sonnet/Haiku, DeepSeek V3).
- **Metric:** retrieval success — correct tender documents obtained per URL, judged by the Evaluator against expected outcomes.
- **Infrastructure:** FastAPI · Celery (4 workers) · Redis · sandboxed execution · MinIO · PostgreSQL.

6 Results

97%	\$0.02	348s	4/75
URLs retrieved (97/100)	total LLM cost	runtime · 4 workers	new / cached LLM calls

MODEL	SUCCESS	LAT.	COST/RUN	GRADE
Gemini 2.5 Flash Lite	83%	4.2s	\$0.0018	A
Gemini 2.5 Flash	83%	4.5s	\$0.0034	A
GPT-4o mini	67%	12.8s	\$0.0027	B
Claude Sonnet	67%	17.7s	\$0.0719	B

Table 1. Per-model extraction quality (top 4 of 10). Without caching, single-shot regeneration reached 87% at \$0.53, quantifying the benefit of the feedback loop and cache.

7 Discussion

- **Caching dominates cost:** only 4 of 79 model calls were new (75 cached), so cost scales with the number of *new domains*, not URLs.
- **Scale ≠ quality:** the smallest model (Gemini 2.5 Flash Lite) matched the largest; pricier models added latency and cost without higher success.
- **Graceful limits:** authentication walls and CAPTCHAs are detected and routed to the CUA agent or flagged for manual action rather than failing silently.

8 Conclusion & Future Work

A self-refining generate-and-verify loop combined with a cost-aware cascade turns fragmented procurement portals into reliable, near-zero-cost document discovery. **Future work:** vision-capable computer-use models, automated regression detection when portals change, and extension beyond Germany to additional jurisdictions.

9 References

1. Madaan et al. *Self-Refine: Iterative Refinement with Self-Feedback*. NeurIPS, 2023.
2. Yao et al. *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR, 2023.
3. Chen et al. *Evaluating Large Language Models Trained on Code*. arXiv:2107.03374, 2021.
4. Zhou et al. *WebArena: A Realistic Web Environment for Building Autonomous Agents*. ICLR, 2024.
5. Schick et al. *Toolformer: Language Models Can Teach Themselves to Use Tools*. NeurIPS, 2023.

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